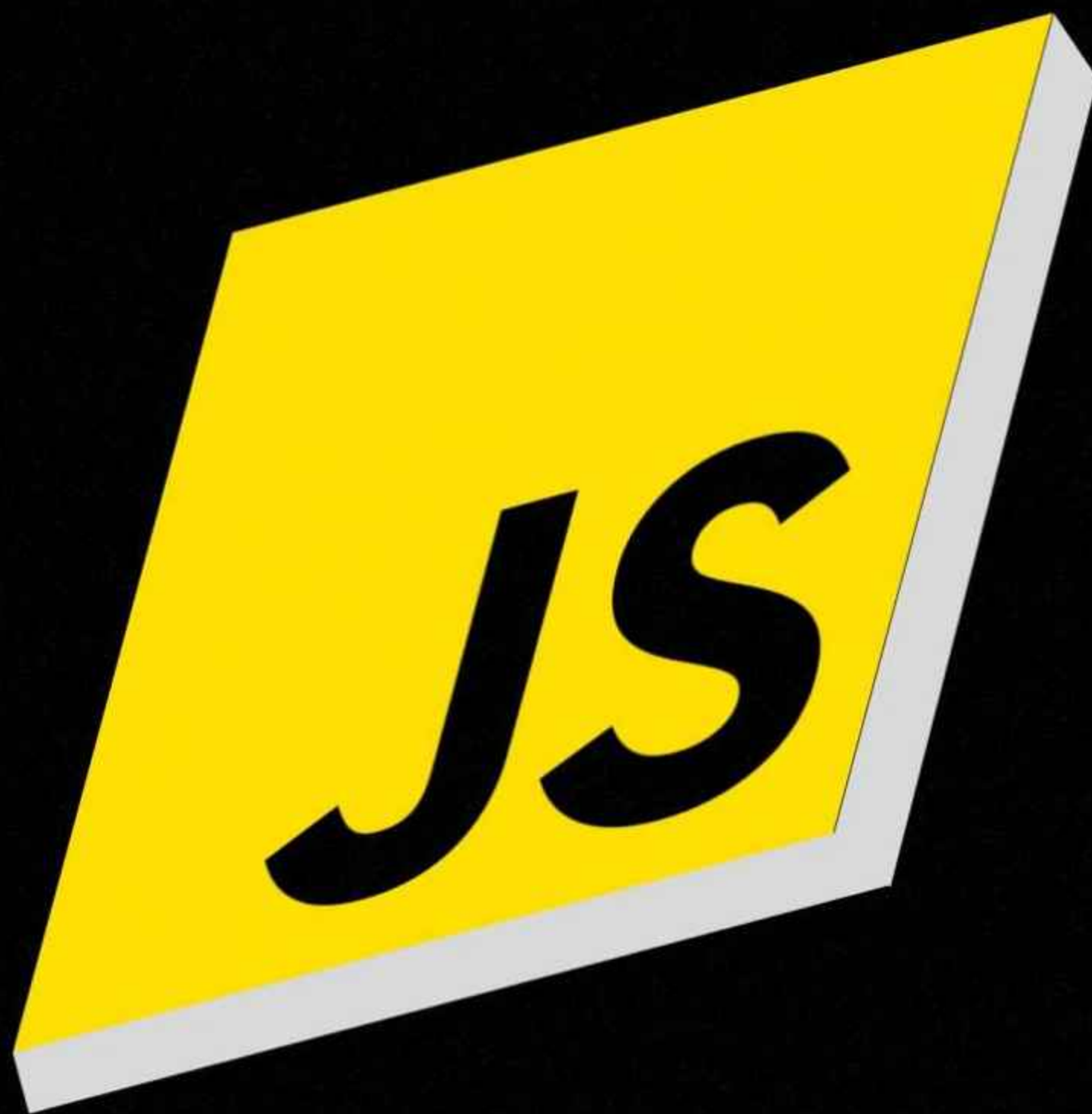




Math object Explained

INCLUDE 8 MOST USEFUL
MATH OBJECT METHODS



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WHAT IS THE MATH OBJECT?

The Math object is a built-in JavaScript object that provides you with methods to perform mathematical operations.

This object can be used directly in your code whenever you need it.

For example, you need to round a floating number:

```
const myNumber = 2.4677;  
  
console.log(Math.round(myNumber)); // 2
```

THESE ARE 8 MOST USED MATH OBJECT METHODS



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1. **random()**

Returns a random number between 0 and 0,999.

You can manipulate the result to upscale the random number range

```
console.log(Math.random()); // 0.18983818813478948
console.log(Math.random()); // 0.9374821802682687
console.log(Math.random()); // 0.9814661909601377

// Generate a random number between 10 and 20 (inclusive)
// Change min and max here as you see fit
const min = 10;
const max = 20;
const randomNumber = Math.floor(Math.random() * (max - min + 1)) + min;
```

2. **round(x)**

Rounds your number to the nearest integer
(can be up or down)

```
// Rounds up when fractional is 0.5 or higher
console.log(Math.round(0.5)); // 1

// Rounds down when fractional is lower than 0.5
console.log(Math.round(2.49999)); // 2
```



3. **ceil**(x)

Rounds up your number, regardless of the floating value

```
console.log(Math.ceil(0.5)); // 1  
console.log(Math.ceil(2.49999)); // 3
```

4. **floor**(x)

Rounds down your number regardless of the floating value

```
console.log(Math.floor(0.5)); // 0  
console.log(Math.floor(2.49999)); // 2
```



5. **pow**(x, y)

Returns **X** raised to a power of **Y**

```
console.log(Math.pow(5, 2)); // 25 (5*5)
```

```
console.log(Math.pow(4, 3)); // 64 (4*4*4)
```

6. **sqrt**(x)

Returns the square root of **X**

```
console.log(Math.sqrt(25)); // 5
```

```
console.log(Math.sqrt(64)); // 8
```



7. **min**(value1, ..., valueN)

Returns the **smallest number** from the arguments you passed to it

```
console.log(Math.min(10, 0, 5)); // 0

console.log(Math.min(-3, -1, -10)); // -10

// If you want to pass an array, use the spread operator
const myArray = [20, 30, 40];

console.log(Math.min(...myArray)); // 20
```

8. **max**(value1, ..., valueN)

Returns the **largest number** from the arguments you passed to it

```
console.log(Math.max(10, 0, 5)); // 10

console.log(Math.max(-3, -1, -10)); // -1

// If you want to pass an array, use the spread operator
const myArray = [20, 30, 40];

console.log(Math.max(...myArray)); // 40
```

